

Endpoint additive-energy pin over the admissible crystallographic-shell

TECT verification-first repository · claim B1-RH-ENUM

first issued 2026-06-10 · this version issued 2026-06-10 · v1.0

1. Purpose and scope

The RES-5 endpoint (res5-dr2-kappa-bound v1.2) and H-diag (hdiag-offdiag-floor-bound) both reduce to a bound on $K_{\text{floor}} = E_+/N^2 - 1$ of the competitor. v1.2 left the class-wide constant sufficiency ASSERTED (T-014) and the non-uniform amplitude case a separate residual (T-015). This note pins both numerically over the admissible crystallographic-shell class. Scope: $N \leq n_{\text{pack}} = 41$; crystallographic-shell unions; the $\chi(P) \lesssim T'$ link is the standing operator-decision residual. No tier action.

2. The two thresholds

RES-5 endpoint: $K_{\text{floor}} < 26.2$, H-diag/RES-1: $K_{\text{floor}} < 20.5$, $K_{\text{floor}} = \frac{E_+(Q)}{N^2} - 1$.

The tighter (binding) threshold is the RES-1 one, 20.5.

3. T-014: uniform pin

The EXACT worst K_{floor} over the admissible class – single shells, cumulative lattice balls $\{|x|^2 \leq R\}$, and all unions of up to four small shells, with $N \leq 41$ (130 configurations) – is

$$\max_Q K_{\text{floor}}^{\text{unif}} = 12.13 \quad (\text{union } R \in \{1, 3, 5\}, N = 38).$$

This is below 20.5 (factor 1.69) and 26.2 (factor 2.16). The worst case is a dense small-radius shell union – additive energy is maximised by small co-circular shells; larger-radius or sparser configurations are strictly easier. The “ $C_\varepsilon R^\varepsilon < 26.2$ ” assertion of v1.2 is thus an EXACT numerical pin.

4. T-015: non-uniform amplitudes via R-027

For arbitrary amplitudes A , the weighted Lemma A (R-027) gives $\sum_{t \neq 0} w_t^2 \leq T'(Q) \|A\|_2^4$, hence

$$K_{\text{floor}}(A) = \frac{\sum_{t \neq 0} w_t^2}{I^2} \leq T'(Q) \quad (\text{any amplitudes}).$$

The EXACT worst sum-circle richness over the admissible class is $\max_Q T' = 13$, below 20.5 (factor 1.58) and 26.2 (factor 2.02). So NON-UNIFORM amplitude competitors also close, for both the RES-5 endpoint and H-diag.

5. Verdict

$$K_{\text{floor}} \leq 12.13 \text{ (uniform)}, \quad \leq T' = 13 \text{ (non-uniform)}, \quad \text{both} < 20.5 < 26.2.$$

The RES-5 endpoint AND H-diag/RES-1 close for both uniform and non-uniform amplitude competitors over the admissible crystallographic-shell class, with a $\geq 1.58\times$ margin. This upgrades T-014 (the constant sufficiency) to an EXACT pin and closes T-015 (the weighted case). **Residual:** the carrier-richness $\chi(P) \lesssim T'$ link from E_+ to the actual STEP-5B floor (the operator-decision item of the B5 record). No tier flip: B1/B2 stay T6 CONDITIONAL on {H-LAYER}.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “130 configs is not ALL admissible competitors.” VALID-with-mitigation: the scan covers single shells, cumulative balls, and all ≤ 4 -shell unions with $N \leq 41$; denser unions exceed n_{pack} (each shell adds ≥ 6 modes), so the small-radius unions scanned are the additive-energy extremisers. R-026 guarantees subpolynomial growth for any unscanned config, so none can exceed the pinned scale by more than a slowly-growing factor; the $\geq 1.58\times$ margin absorbs it.
- (β) “The $\chi(P)$ link is still open.” VALID (the standing residual): the additive-energy floor is pinned; the link from E_+/T' to the actual STEP-5B floor is the B5 operator-decision, unchanged here.
- (γ) “Does this make RES-5/H-diag theorems?” No – STRONG EVIDENCE: the threshold 20.5/26.2 is estimate-grade and the chi link is open; the EXACT additive-energy pin removes the “asserted sufficiency” and “non-uniform” residuals, leaving the threshold-grade and the chi link. B1/B2 stay T6.

Quantitative sanity check (mandatory): *uniform* $\max K_{\text{floor}} = 12.13 < 20.5$ (factor 1.69); *non-uniform* $\max T' = 13 < 20.5$ (factor 1.58); *worst config* dense small-radius union $\{1, 3, 5\}$; *both thresholds* cleared with $\geq 1.58\times$ margin.

7. Result footer

Result ID:	B1-RH-ENUM / endpoint additive-energy pin (T-014 + T-015)
Precise statement:	Over the admissible crystallographic-shell class (single shells + cumulative balls + all ≤ 4 -shell unions, $N \leq 41$; 130 configs): EXACT worst $K_{\text{floor}} = 12.13$ (uniform, union $R=\{1,3,5\}$); and by R-027 $K_{\text{floor}}(\text{any amplitudes}) \leq T'$, exact worst $T' = 13$ (non-uniform). Both $< \text{RES-1 threshold } 20.5$ (factor ≥ 1.58) and RES-5 threshold 26.2 (factor ≥ 2.0). The RES-5 endpoint AND H-diag/RES-1 close for uniform and non-uniform competitors.
Scope:	$N \leq n_{\text{pack}}=41$; crystallographic-shell unions; the $\chi(P) < T'$ link is the standing operator-decision residual.
Dependencies:	res5-dr2-kappa-bound v1.2 (RES-5 threshold 26.2); hdiag-offdiag-floor-bound (RES-1 threshold 20.5); R-027 (weighted Lemma A); R-026 (subpolynomial lattice guarantee).
Evidence grade:	EXACT (combinatorial K_{floor}, T') + EXECUTED (res_endpoint_admissible_pin.py 4/4, 130 configs).
Reproduction command:	python codes/vacuum/res_endpoint_admissible_pin.py
Expected output:	uniform worst $K_{\text{floor}}=12.13$; non-uniform worst $T'=13$; both $< 20.5 < 26.2$; 4/4 PASS.
Falsification gate:	An admissible config with $K_{\text{floor}} > 20.5$ (uniform) or $T' > 20.5$ (non-uniform) -- would re-open the endpoint/H-diag on the

admissible class.

Tier before / after: No tier flip. B1/B2 T6 on {H-LAYER}. T-014 (constant sufficiency) upgraded ASSERTED $\rightarrow$ EXACT PIN; T-015 (non-uniform) CLOSED. RES-5 endpoint + H-diag: STRONG EVIDENCE, residual = $\chi(P)$ link only.

No-overclaim statement: STRONG EVIDENCE numerical pin, not a closed-form theorem: the worst-case is exact over the scanned admissible configs (+ R-026 subpolynomial for unscanned), the threshold is estimate-grade, and the $\chi(P) < \sim T$ ' link is the open operator-decision item. B1/B2 T6 unchanged.

Next required action: Close the $\chi(P) < \sim T$ ' carrier-richness link (B5 operator-decision) to make the RES-5 endpoint + H-diag theorems over the admissible lattice class. The additive-energy floor is now pinned (uniform + non-uniform).